

$^1\text{H}(^{34}\text{Ar}, \text{P}') \quad \text{2001Kh17}$

[2001Kh17](#): a 47-MeV/nucleon ^{34}Ar beam was produced via the projectile fragmentation of a 95-MeV/nucleon ^{36}Ar primary beam impinging on a ^{12}C target at GANIL. The secondary target was CH_2 . The elastic and inelastic protons were detected using the MUST Si-Si(Li)-CsI telescope and heavy nuclei were analyzed by the SPEC spectrometer. Measured $\sigma(E_p, \theta)$. Deduced levels, J, π , and deformation parameters from Coupled-Channel Born Approximation (CCBA) using the ECIS code and TAMURA-DWBA analyses of the measured $\sigma(\theta)$. Also see [2000KhZX](#), [2001B117](#), [2003La13](#) for more discussions on the same data.

 ^{34}Ar Levels

E(level)	J π [†]	Comments
0	0 ⁺	Ratio of quadrupole matrix elements $M_n/M_p/(N/Z)=1.35 \ 28$ from phenomenological analysis (2001Kh17) and $0.99 \ 17$ from the QRPA transition densities normalized to the data in 2001Kh17 .
$1.9 \times 10^3 \ 2$	2 ⁺	$\beta_2=0.27 \ 2$ (2001Kh17)
$4.5 \times 10^3 \ 2$	3 ⁻	$\beta_3=0.39 \ 3$ (2001Kh17)

[†] From CCBA and DWBA analyses of measured $\sigma(\theta)$ in [2001Kh17](#).